

Ilyes Ouldsaada

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EDUCATION

Massachusetts Institute of Technology | Cambridge, MA

May 2029

Bachelor of Science in Computer Science and Computer Engineering – GPA: 4.7/5.0

Coursework: Intro to Algorithms, Modeling with Machine Learning for Computer Science, Fundamentals of Programming, Mathematics for Computer Science, Multivariable Calculus, Building Large Language Models from Scratch

SKILLS + TECHNICAL TOOLS

Languages: Python, Java, Javascript, Bash, HTML/CSS, C#

Technologies: PyTorch, Tensorflow/Keras, Scikit-learn, Pandas, Numpy, Matplotlib, Scipy, Git, Jupyter, Pydantic, Flask, PIL

EXPERIENCE

Data Strategy Intern | Fidelity Investments Strategic Advisors

Boston, MA | Jun 2026 – Aug 2026

- Architected an in-house Due Diligence **analytics site** that automated financial document analysis using **Python, HTML, CSS, and JavaScript**, cutting manual review time by **50%** for analysts while providing senior leadership with risk audit trails.
- Engineered an automated data pipeline to parse JSON from unstructured institutional Due Diligence Questionnaires; architected a **Retrieval-Augmented Generation (RAG)** system utilizing **vector databases** to efficiently **query document embeddings** and structure qualitative model outputs, ensuring the model had domain-accurate knowledge.
- Created a robust peer-benchmarking engine using **Isolation Forests, Multivariate Gaussian Copulas, and Kernel Density Estimation (KDE)** to model high-dimensional joint probability distributions of textual document risk attributes — including domain-specific **sentiment signals** extracted via a **fine-tuned FinBERT** model; validated risk thresholds and structural document outliers via Tukey's Fences and multi-fold **cross-validation**.

Undergraduate Researcher | MIT-Harvard Biomedical Center

Cambridge, MA | Dec 2025 – March 2026

- Engineered deep learning workflows for automated coronary plaque image segmentation and OCT/OAC architecture benchmarking in Python; **collaborated closely** with clinical mentors to interpret quantitative findings and adapt our modeling strategy to overcome ambiguous ground truth labels and **class distributional irregularities**.
- Designed and systematically benchmarked **14 deep learning architectures** to evaluate relative **signal extraction performance** between two optical imaging modalities, implementing automated training pipelines.
- Executed **structured parameter sweeps** and multi-objective optimization across UNet architectures; optimized cross-validated loss metrics improve **model generalization** under severe class imbalance.

PROJECTS

Frugal Sheets: Reactive Spreadsheet Engine | *Python, Graph Algorithms, Object-Oriented Design*

- Engineered a live dependency-graph engine for a spreadsheet application, recursively parsing cell formulas to extract references and maintaining a **directed graph** of dependent cells that updates in real time as formulas change.
- Implemented a depth-first **topological sort** to recompute only downstream-affected cells in correct order on every edit, with cycle detection to reject circular formulas; extended the graph with **Region** nodes for row/column-level aggregates.

KinoML: Adaptive Film Recommendation System | *Python, Pytorch, Numpy, Object Oriented Programming*

- Designed, implemented, and evaluated multi-step **meta-learning regressors** from scratch in PyTorch, leveraging **episodic task sampling** to enable rapid model adaptation to unseen user profiles under **extreme data scarcity**.
- Created a custom **SimCLR self-supervised learning pipeline** with **stochastic** masking, dropout, and noise augmentations to project unstructured, real-world film and user data attributes into continuous, learned vector representations.